Aditi Pore

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SUMMARY

Third-year Electronics and Telecommunication Engineering student at COEP Technological University pursuing a Minor in AI & ML, with hands-on experience in embedded systems, analog circuit design, hardware-software integration, and AI-enabled embedded solutions. Technical Secretary of the COEP Amateur Radio Club (VU2COE) and licensed Amateur Radio Operator, experienced in delivering technical workshops, coordinating cross-functional teams, and communicating complex engineering concepts to diverse audiences. Passionate about semiconductor technology, application engineering, technical sales, customer engagement, and solving real-world engineering challenges through collaborative problem solving.

EDUCATION

COEP Technological University — B.Tech, Electronics and Telecommunication Engineering; Minor in AI and ML (2024 to 2028)
JEE Mains: 96.98 percentile | MHT-CET: 99.65 percentile | CGPA: 8.3

TECHNICAL SKILLS

Programming Languages: Python, C, SQL
Embedded Systems: ESP32, ESP32-CAM, 8051, MPU6050 IMU
Communication Protocols: I2C, I2S, UART, Bluetooth Low Energy (HID)
Analog and Hardware: Op-amps, instrumentation amplifiers, active filters, signal conditioning
Simulation and Test Equipment: Proteus, MATLAB, RIGOL digital storage oscilloscope
Software and Data Tools: Groq LLaMA API, Google TTS, MySQL
Technical & Professional Skills: Technical Communication, Technical Presentations, Customer Engagement, Cross-functional Collaboration, Problem Solving, Troubleshooting

PROJECTS

AI Vision Assistant for the Visually Impaired
ESP32-CAM, Groq LLaMA API, Google TTS, I2S

- Built an ESP32-CAM device that captures a user's surroundings, sends frames to the Groq LLaMA model for scene analysis, and converts the description to speech over I2S audio.
- Implemented Base64 image encoding and REST API calls between the microcontroller and cloud inference, and added an OLED status display and low light flash support.

Seven-Stage Analog EMG Signal Processing Circuit
TL071 Op-Amps, Instrumentation Amplifier, Analog Front End

- Designed a seven-stage analog front end for surface EMG signals: an instrumentation amplifier (gain 201), band-pass filtering, a 50 Hz twin-T notch filter, and an envelope detector.
- Achieved approximately 10,000 V/V of gain with a clean 0 to 5V DC output suitable for prosthetic motor control.

KHN State-Variable Audio Filter — *Team Cornerstone Project*
Op-Amp Based Active Filter, Bode Plot Validation

- Collaborated with two teammates to design a Kerwin-Huelsman-Newcomb active filter at 1 kHz with simultaneous low-pass, band-pass, and high-pass outputs and independent control of Q and frequency.
- Validated frequency response against theoretical Bode plots on a RIGOL digital storage oscilloscope.

Motion-Controlled Air Mouse (IMU Based)
ESP32, MPU6050, Bluetooth Low Energy HID

- Built a motion-controlled air mouse using an ESP32 and MPU6050 IMU, translating hand motion into wireless cursor control through a custom BLE HID profile.
- Added debounced click buttons, a scroll mode, and a battery-powered portable design.

LEADERSHIP AND INVOLVEMENT

- **Technical Secretary, COEP Amateur Radio Club (VU2COE)**, June 2026 to present — Co-organizes antenna and other workshops and fox hunts, and administers government amateur radio licensing exams for club members. Coordinate club events end-to-end and represent the club in interactions with external organizations, building professional relationships.
- Volunteer, COEP Amateur Radio Club (VU2COE), 2024 to 2026
- NGO Volunteer, Azad Nayakawadi Foundation, Kolhapur
- Volunteer, Mindspark 2024 — COEP Technological University's flagship technical festival

LICENSES

Amateur Radio Operator License (Restricted Grade), Ministry of Communications, Government of India — VU39AP

EXPERIENCE

Database Intern — Softwin Infotech *6 weeks, concluded July 2026*

- Managed and troubleshot production databases in SQL and MySQL, strengthening query writing, debugging, and data driven problem solving in a live business environment.
- Communicated findings and solutions clearly with the team, strengthening technical communication, attention to detail, and business awareness.